StructEye LoRa Communication Protocol

The StructEye system consists of one master node (base station) and multiple slave sensor nodes placed at critical points across a building structure. Each slave node is equipped with an MPU6050 sensor (accelerometer + gyroscope), an STM32F446RE microcontroller, and a LoRa SX1278 radio module.

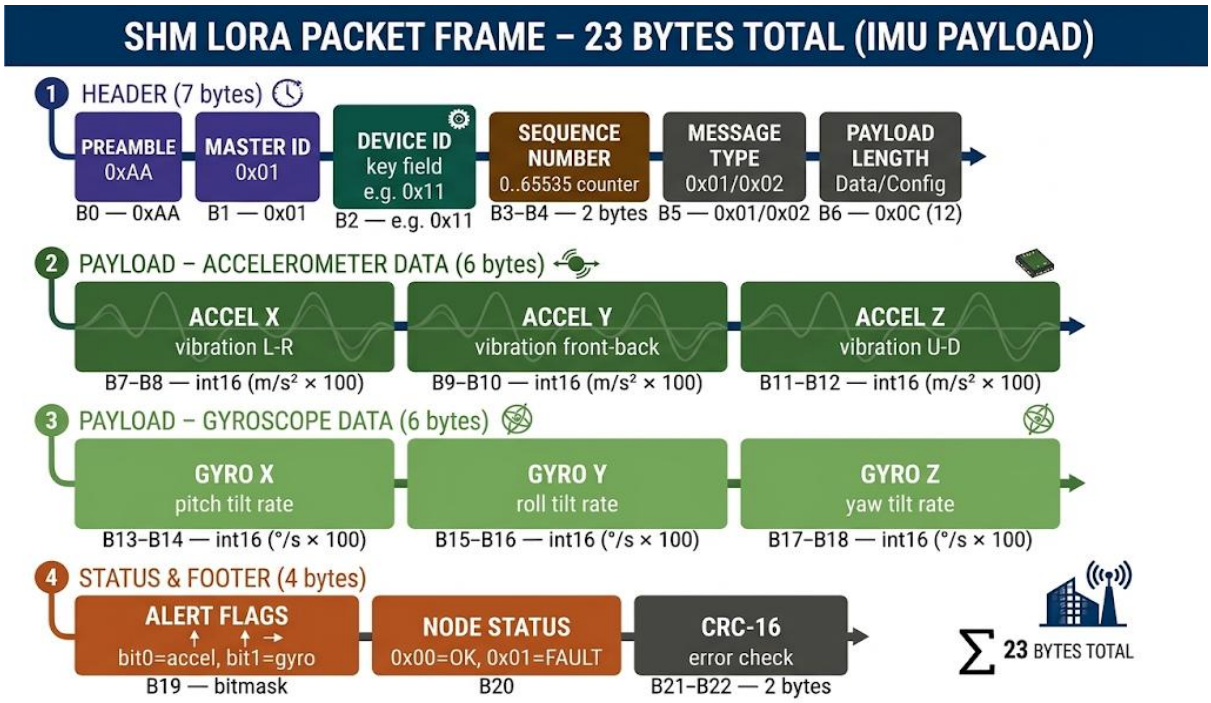
Communication happens over LoRa at 433 MHz using a custom structured frame format. The master controls all communication — it polls each slave by sending a packet containing the slave's unique Device ID. Only the addressed slave responds with its sensor data.

LoRa Packet Frame Structure

When the master wants to collect data from a slave, it does not simply transmit raw sensor values. Instead, it organises the data into a structured sequence of bytes called a **frame**.

The StructEye frame is 23 bytes long and is always structured in the same order, so both the master and the slave know exactly what each byte means without any confusion.

Every packet transmitted over LoRa in the StructEye system follows a fixed 23-byte frame structure. This ensures both master and slave nodes can always interpret any received packet without Confusions.



Byte 0 - Preamble (0xAA): A fixed sync byte. Every **StructEye** packet starts with 0xAA. The LoRa module checks this first to confirm the incoming data is a valid system packet and not random noise.

Byte 1 - Master ID (0x01): Identifies the sender. The base station is always 0x01, which allows the system to scale if more master units are added later.

Byte 2 - Device ID: The address of the specific slave node that should respond (e.g., 0x11 for Node B). While all nodes receive the signal, only the one with a matching ID acts on it. 0xFF is used as a broadcast ID for all nodes.

Bytes 3–4 - Sequence Number: A 16-bit counter (0–65535) that increments with every packet. It prevents processing duplicate wireless packets and allows the master to detect if a specific packet (e.g., #3) was lost in transmission.

Byte 5 - Message Type: Defines the packet's purpose. 0x01 is a poll request (master asking for data); 0x02 is a data response (slave sending sensor readings).

Byte 6 - Payload Length (0x0C): Specifies the size of the sensor data. In **StructEye**, this is fixed at 12 bytes (6 for accelerometer and 6 for gyroscope).

Bytes 7–12 - Accelerometer Data (X, Y, Z): Six bytes carrying vibration readings. Each axis is a signed 16-bit integer. Values are multiplied by 100 before packing (e.g., 12.53 m/s^2 becomes 1253) to avoid sending slow floating-point numbers.

Bytes 13–18 - Gyroscope Data (X, Y, Z): Six bytes carrying tilt readings. Like the accelerometer, values are multiplied by 100 and stored as signed 16-bit integers (e.g., 1.23 °/s becomes 123).

Byte 19 - Alert Flags: A bitmask for warning conditions.

Bit 0 (0x01): Accelerometer exceeded threshold.

Bit 1 (0x02): Gyroscope exceeded threshold.

Bit 2 (0x04): Sensor communication error.

Byte 20 - Node Status: The final health verdict. 0x00 means the structure is OK; 0x01 means a FAULT was detected. This is the first byte the master checks upon receipt.

Bytes 21–22 - CRC-16 (Error Check): A mathematical checksum of all previous bytes. The receiver recomputes this value; if it doesn't match the transmitted CRC, the packet is discarded as corrupted, ensuring data integrity.

B0	B1	B2	B3-4	B5	B6	B7-12	B13-18	B19	B20	B21-22
Sync	MID	DID	SEQ	TYPE	LEN	ACCEL	GYRO	FLAG	STAT	CRC

How the Device ID Selects a Slave

This is the core mechanism of the StructEye communication system. The following describes exactly what happens when the master sends a poll request to Node B.

Step 1. The master builds a frame with Device ID = 0x11 at byte B2 and hands it to the SX1278 LoRa module. The SX1278 broadcasts this packet over the air at 433 MHz.

Step 2. All three slave nodes — Node A, Node B, and Node C — receive the packet simultaneously. LoRa is a broadcast medium, meaning there is no concept of a private direct link. Every node hears everything.

Step 3. Each slave's STM32 microcontroller reads byte B2 from the received frame and compares it with its own Device ID stored in the firmware.

Step 4. Node A reads 0x11 and compares with its own ID 0x10. They do not match. Node A discards the packet immediately and returns to listening mode without doing anything further.

Step 5. Node C reads 0x11 and compares with its own ID 0x12. They do not match. Node C discards the packet and returns to listening mode.

Step 6. Node B reads 0x11 and compares with its own ID 0x11. They match. Node B now activates its MPU6050, reads the vibration and tilt data, builds a response frame, and transmits it back to the master.

Step 7. The master receives Node B's response, checks the sequence number to confirm it matches the request, reads the sensor values from bytes B7–B18, checks the alert flags in B19, and reads the node status in B20 to decide whether to display OK or NOT OK.

Summary

The StructEye LoRa communication system uses a fixed 23-byte frame where the Device ID byte selects which slave responds, making all other nodes ignore the packet. The sequence number prevents duplicates and detects lost packets. The MPU6050 provides six axes of data - three for vibration (accelerometer) and three for tilt (gyroscope) - which are converted from raw register values, scaled by 100 to preserve decimal precision, and packed as signed 16-bit integers into the payload. The slave evaluates its own data against thresholds before transmitting and sets the alert flags and node status bytes accordingly, so the master can immediately determine the structural health condition at each monitored point without any additional processing.